



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE



Text Analytics with Language Technology and Data Analysis Laboratory Resources: an introduction to free, open-source interactive resources for linguists

Martin Schweinberger



LADAL:

Language Technology and Data Analysis Laboratory

Collaborative, cloud-based workbench environment, bringing together users and providers of data and text analytics tools. It will support researchers transitioning to code-based text analysis, with the resultant benefits of flexibility, reproducibility and reuse.

Aim to **develop computational HASS infrastructure**

- **Collation of projects** (UQ: LADAL | USyd: SCL)
(diversity of audiences, needs, and approaches)
- **Development of resources**
(> 30 tutorials, 15 interactive notebooks)
(websites, tools, static and interactive notebooks)
- **Upskilling | engagement**
(workshops, seminars, trainings)
- **Continuing support** (consultation and 1:1 support)
- Generating **vectors** (educators and fellows)



Language Technology and Data Analysis Laboratory (LADAL)

LADAL: <https://ladal.edu.au>

Free, open-source eResearch support infrastructure for computational HASS in the UQ School of Languages and Cultures

Aim

- provide (notebook-based) resources for computational HASS (website tutorials and case studies, interactive notebooks, links, information, etc.)
- organize events and community building (webinars, workshops, etc.)

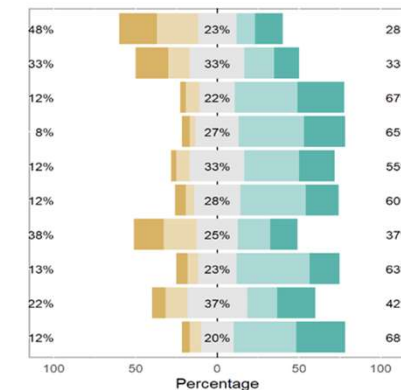
People

- Directors: Martin Schweinberger & Michael Haugh
- Contributors: Sam Hames, Ben Foley plus various international experts



Michael Haugh
(co-director of
LADAL)



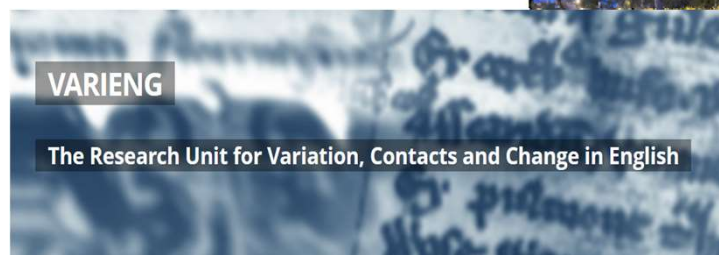
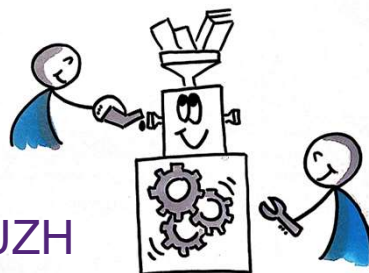


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International Collaborations

- Text Crunching Center (TCC) at UZH
- Varieng at UHel
- Syney Corpus Lab
- AcqVA Aurora at UiT



(sharing of resources, reciprocal promotion, and assistance)



AcqVA Aurora Lab

Sydney Corpus Lab

Discover the Power of
Computer-based Text
Analysis

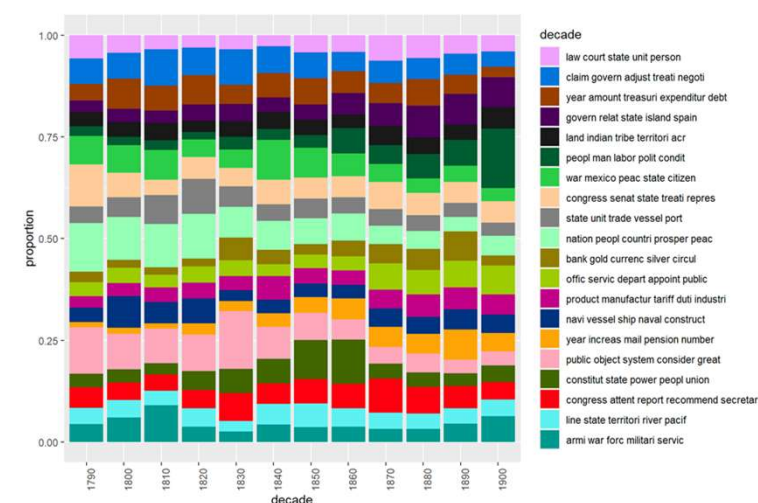
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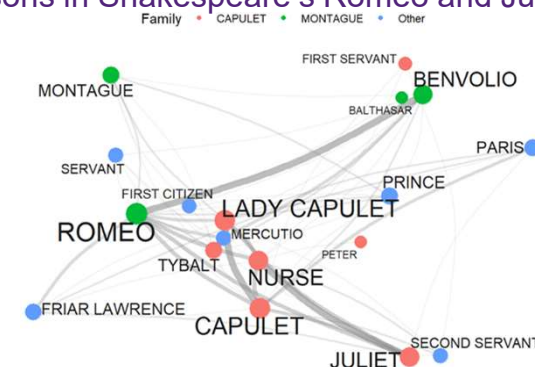
Distribution of topics in US State of the Union Addresses over time

What we hope to achieve?

- **Improve transparency and quality** by showcasing how to produce reproducible workflows in R/RStudio)
- Enable researchers to pursue **new pathways** by using innovative methods and new types of data
- Improve **data management**, assist in making workflows **tidier**, more **transparent** and **more efficient**.
- Provide an **infrastructure for acquiring computational skills** (relevant for academia | employability for graduates)
- Showcase **comp. methods** can be used across **HASS disciplines**



Network of persons in Shakespeare's Romeo and Juliet

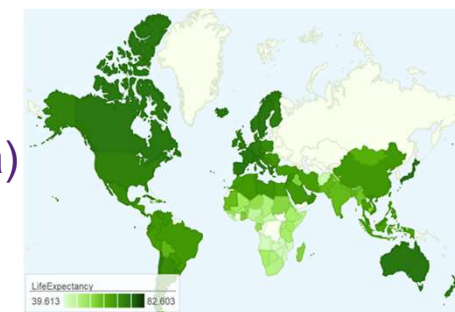
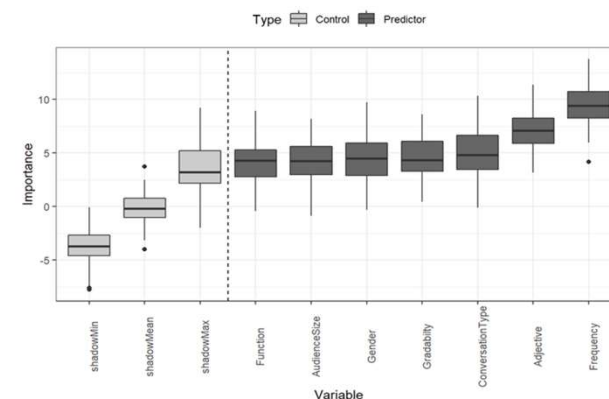
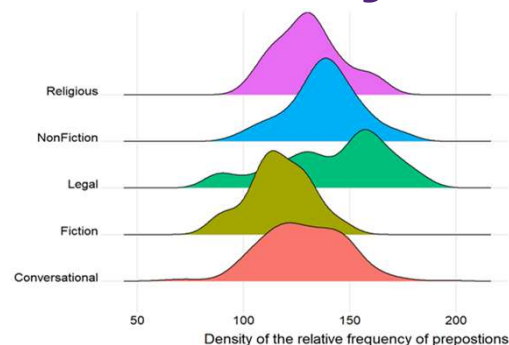


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Key measure so far:

- 40 tutorials, case studies, and how-tos
- 15 interactive Jupyter notebooks (on Binder.org and Google Colab)
- More than **450,000 page views**
- More than **250,000 different users**
- LADAL resources used **all around the world** (from university courses on quantitative methods in Norway to dashboards for medical school in California)
- **diverse users from various fields** (from linguists to psychologists to artists)



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PRACTICE

- Go to ladal.edu.au
- Click on the tab **TUTORIALS**
- Go to **Text Analysis**
- Click on **Practical Overview of Selected Text Analytics Methods**
- Click on the **Binder.org** link



Tutorials

Created by LADAL contributors
2022-11-08



5 Text Analytics

Introduction to Text Analysis

This tutorial introduces Text Analysis, i.e. computer-based analysis of language data or the (semi-)automated extraction of information from text.

Practical Overview of Selected Text Analytics Methods

This tutorial showcases some basic but useful methods for text analysis and serves as a practical overview or introduction to Text analytics and distant reading.

Concordancing (keywords-in-context)

This tutorial introduces how to find words or phrases in text and display concordances, a so-called keyword-in-context (KWIC) display, with R.



The entire R Notebook for the tutorial can be downloaded [here](#). If you want to render the R Notebook on your machine, i.e. knitting the document to html or a pdf, you need to make sure that you have R and RStudio installed and you also need to download the **bibliography file** and store it in the same folder where you store the Rmd file.



Click this link to open an interactive version of this tutorial on [MyBinder.org](#).

This interactive Jupyter notebook allows you to execute code yourself and you can also change and edit the notebook, e.g. you can change code and upload your own data.